

Emo3D: Metric and benchmarking dataset for 3D facial expression generation from emotion description

Anonymous ACM Submission

ABSTRACT

Existing 3D facial emotion modeling have been constrained by limited emotion classes and insufficient datasets. This paper introduces “Emo3D”, an extensive “Text-Image-Expression dataset” spanning a wide spectrum of human emotions, each paired with images and 3D blendshapes. Leveraging advanced Language Models (LLMs), we generate a diverse array of textual descriptions, facilitating the capture of a broad spectrum of emotional expressions. Using this unique dataset, we conduct a comprehensive evaluation of language model fine-tuning and CLIP-based models for 3D facial expression synthesis. We also introduce a new evaluation metric for this task to more directly measure the conveyed emotion. Our new evaluation metric, Emo3D, demonstrates its superiority over Mean Squared Error (MSE) metrics in assessing visual-text alignment and semantic richness in 3D facial expressions associated with human emotions. “Emo3D” has great applications in animation design, virtual reality, and emotional human-computer interaction.

KEYWORDS

3D Facial Expression Generation, Visual-Language Learning, Multimedia, Benchmark

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1 INTRODUCTION

Automatic translation of character emotions into 3D facial expressions is an important task in digital media, owing to its potential to enhance user experience and realism. Facial Expression Generation (FEG) has a wide range of applications across various industries, including game development, animation, film production, and virtual reality. Previous studies in this domain have primarily focused on generating facial expressions for 2D or 3D characters, often relying on a limited set of predefined classes [20] or driven by audio cues [7, 15]. However, there is a growing need for better control in the generation of complex and diverse human facial expressions.

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Recent studies [10, 23, 24] have made notable progress in this area through the use of text prompts, offering a more direct approach to address the challenge of limited control that has been prevalent in earlier works [7, 15, 20].

The primary issue with recent works using text prompts is (i) their limited focus on textual descriptions of emotions. Many studies have not deeply explored emotional context. These studies have not offered a comprehensive solution that integrates both textual descriptions and 3D facial expression generation, creating a noticeable gap in the field [23, 24]. Moreover, there is (ii) a scarcity of datasets containing emotional text alongside corresponding 3D facial expressions, impeding the development and training of facial expression generation (FEG) models for practical applications [10, 23, 24]. Additionally, (iii) the absence of reliable benchmarks and standardized evaluation metrics in this research area further complicates the assessment of FEG models.

Contributions: This paper tackles key challenges in Facial Expression Generation, focusing on generating expressions from textual emotion descriptions. Our contributions towards addressing the gaps in the field of Facial Expression Generation are as follows: **(i) Emo3D-dataset:** We present the Emo3D-dataset, specifically developed to bridge the gap between textual emotion descriptions and 3D facial expression generation. This dataset provides a rich compilation of annotated emotional texts alongside matching 3D expressions for effective training and assessment of FEG models. **(ii) Baseline Models:** We propose baseline FEG models as benchmarks for future research in this field. These models offer a reference point for evaluating new advancements and assessing progress. **(iii) Evaluation Metric:** To address the absence of standardized evaluation metrics in FEG, we introduce a new metric designed for the unique challenges of capturing the complexities of human emotions.

2 RELATED WORK

Audio-based emotion extraction: Facial expression generation methods often utilize audio data, leveraging the semantic, tonal, and expressive qualities of voice for 3D generation. ‘Audio-driven Facial Animation’ [7] learns to map audio waveforms to 3D facial coordinates, identifying a latent code for expression variations beyond audio cues. ‘EmoTalk’ [15] focuses on creating 3D facial animations driven by speech, aligning expressions with both content and emotion.

CLIP-based baselines: The utility of CLIP’s language-and-vision feature space [16] in text-to-image generation has been highlighted in several works. MotionCLIP [21] leverages CLIP for a feature space that accommodates dual modalities, enabling out-of-domain actions and motion integration into CLIP’s latent space. The 4D Facial Expression Diffusion Model [24] introduce a generative framework for creating 3D facial expression sequences, utilizing a Denoising Diffusion Probabilistic Model (DDPM). The framework consists of two tasks: learning a generative model based on 3D landmark sequences

and generating 3D mesh sequences from an input facial mesh driven by the generated landmarks. The DDPM reverse process can be conditioned on expression labels, text, partial sequences, or a facial geometry. For text control, a Bidirectional Transformer (BiT) is used to output a 512-dimensional vector. Conditioning on text involves maximizing a combination of log-likelihood terms and cosine similarity between the BiT output and textual features extracted with CLIP from the associated text. Also, ExpCLIP [23] is an autoencoder designed to establish semantic alignment among text, facial expressions, and facial images. ExpClip introduces a blendshape encoder to map blendshape weights to an embedding, reconstructed by a decoder. Concurrently, a CLIP text encoder (ϵ_{text}) and text projector (P_{text}) map emotion text into a joint embedding space, with ϵ_{text} frozen during training for enhanced generalization. Similarly, a comparable approach is employed for image information, where an image encoder (ϵ_{img}) is used alongside an image projector (P_{img}), with only P_{img} being fine-tuned. Subsequently, they employ ExpClip for the generation of emotional speech-driven facial animation, wherein the animation generator accommodates both images and natural language prompts for emotion control.

Additionally, [8] introduced CLIPER, a unified framework for both static and dynamic facial expression recognition, utilizing CLIP and introducing multiple expression text descriptors (METD) for fine-grained expression representations, achieving state-of-the-art performance by a two-stage training paradigm which involves learning METD and fine-tuning the image encoder for discriminative features.

Metrics: While numerous metrics exist for 2D image generation, a notable gap persists in effective metrics for 3D facial expression generation. R-precision, used to measure the alignment between input text and output image, was adopted by [3] following the approach in [22]. This involved utilizing a CLIP model fine-tuned on the entire dataset, with R-Precision calculations based on [14].

3 DATASET

We introduce the Emo3d-dataset, an assembly of 150,000 instances. Each instance comprises a triad: textual description, corresponding image, and blendshape scores created as follows:

(i) Emotion descriptions: For generating emotion-specific textual descriptions, we prompt GPT-3.5 [13], with a focus on eight principal emotions for this purpose. Secondly, we again hard-prompt GPT-3.5 language model to derive emotion distributions for these textual elements. This process entailed the construction of specialized prompts for GPT, leading to the generation of eight-dimensional vectors representing distinct emotion distributions, detailed in Figure 1.a.

(ii) 2D Image Generation: Subsequently, we utilize DALL-E 3 [17], a hierarchical image generation model, to create images that align with the generated textual descriptions. Detailed view can be found in Figure 2.

(iii) Blendshape scores estimation: We employ Mediapipe framework [9] to synthesize blendshape scores corresponding to the generated images based on textual descriptions.

Primitive Emotion Faces: Additionally, for intrinsic evaluation purposes, we construct a dataset of primitive emotions comprising singular emotion words, each paired with corresponding images

that portray males and females exhibiting three distinct intensity levels of emotion. Utilizing Mediapipe [9], we subsequently extract blendshape scores for the facial expressions depicted in these images. The emotional distributions associated with these individual words are derived using Emolex [11].

Human Evaluation: Human evaluation was conducted to assess the accuracy of the emotion distribution of the dataset. The results showed a high agreement rate of 81%, indicating that the generated emotion distributions are mostly aligned with human understanding of emotions.

Comparison of Emo3D with existing datasets: Emo3D stands out for its integration of textual, visual, and blendshape modalities, offering a comprehensive representation of emotional expressions. Compared to single-modal datasets [1, 12, 19], Emo3D provides a more holistic representation of emotional expressions. Additionally, With 90,000 images and 60,000 texts, it facilitates emotion recognition in both modalities. Emo3D's emphasis on emotion descriptions distinguishes it from other datasets like TEAD [23] and TA-MEAD[10], which focus on different aspects of facial expression generation. Furthermore, it can be employed for emotion recognition in text and images, thanks to the associated emotion distributions. Table 1.

Dataset	Size	Emotion Distribution	Modalities
AffectNet	440,000	Yes	Image
Emo135	700,000	Yes	Image
CARER	417,000	Yes	Text
TEAD	50,000	No	Text Blendshape
TA-MEAD	-	No	Text Video
Emo3D	Text: 60,000	Yes	Text
	Image: 90,000		Image
	Triple: 150,000		Blendshape

Table 1: Comparison of Emo3D with existing datasets:This table summarizes key attributes of Emo3D alongside established datasets such as AffectNet [12], Emo135 [1], CARER [19], TEAD [23], and TA-MEAD [10].

The Emo3d-dataset shares similarities with other existing datasets, particularly TEAD[23] and TA-MEAD[10], in terms of modality integration and a focus on emotional expressions.

The TA-MEAD [10] dataset, designed for 2D facial expression generation (FEG), provides emotion descriptions for videos, along with Action Unit (AU) [5] intensity annotations for each video. In contrast, our Emo3d-dataset offers a unique perspective by concentrating on textual emotion expressions, corresponding images, and blendshape scores.

The TEAD [23] dataset, designed for 3D FEG, features situation descriptions, our Emo3d-dataset distinguishes itself by emphasizing emotion descriptions. Additionally, our dataset includes a distinctive feature with corresponding images for each text, providing a richer and more comprehensive resource. The Emo3d-dataset, comprising 150,000 samples, stands out significantly in scale when compared to ExpClip, which consists of 50,000 samples.

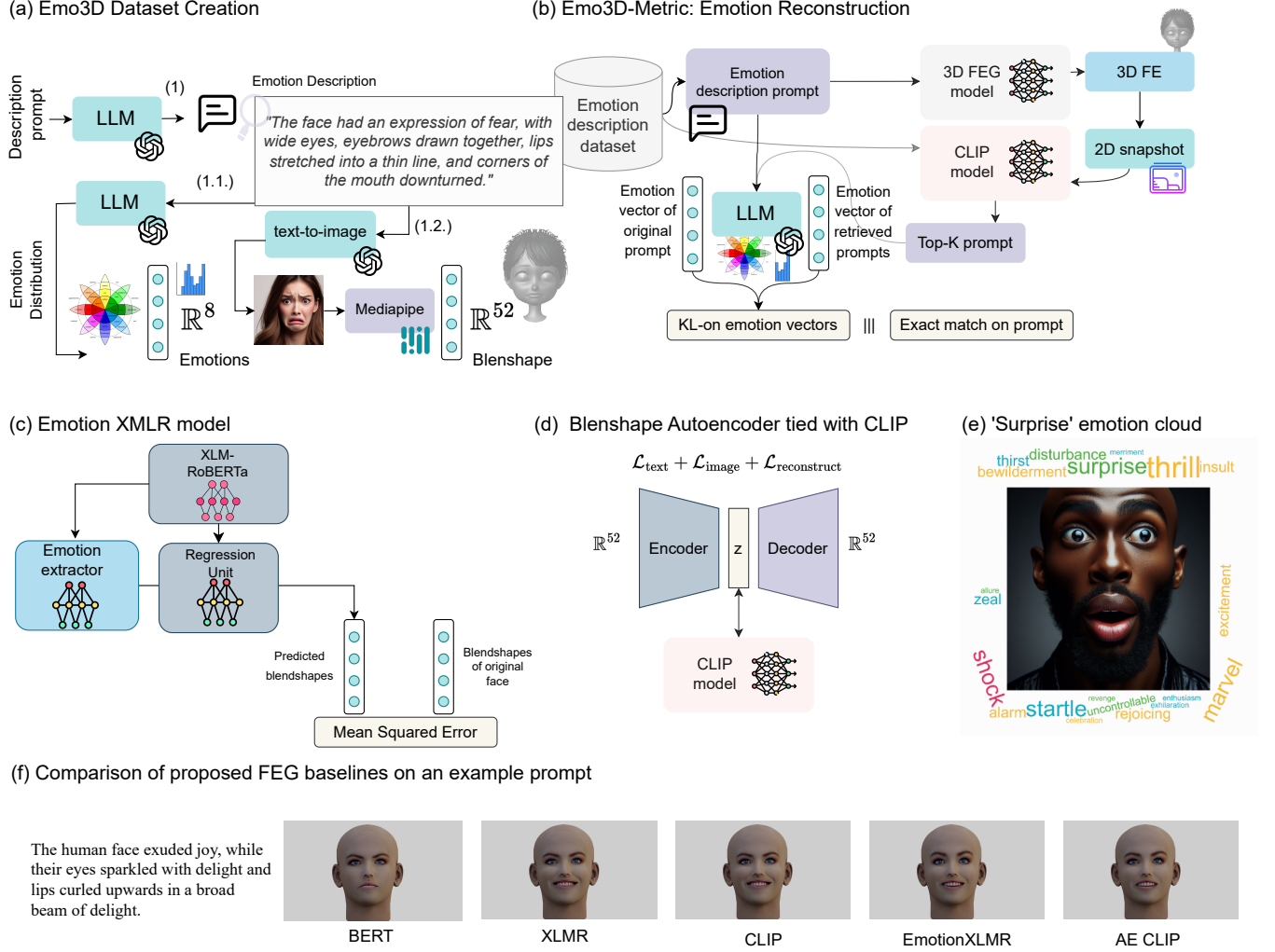


Figure 1: (a) Emo3D-datasets Generation Pipeline: Textual data describing human emotions is initially generated using GPT [13]. We then utilize DALL-E models [17] to synthesize human faces. Each image undergoes face blendshape extraction using MediaPipe [9]. Furthermore, we employ GPT [13] to extract the emotion distribution for each prompt. **(b) Emo3D Metric:** Our methodology entails selecting n prompts with a balanced emotion distribution. We generate facial expressions using a text-utilizing FEG model for a given input prompt. We project the 3D face model onto a 2D image and employ zero-shot CLIP to identify the k nearest text prompts. Subsequently, we compute the Kullback-Leibler (KL) divergence between the emotion distribution of the input text and these k prompts. **(c) Emotion-XLM:** The model uses emotion ground truth to predict facial blendshapes. An Emotion Extractor guides the Regression model with the Teacher-Forcing technique at a 50% ratio. Both units are trained via Mean Squared Error (MSE) loss. **(d) Autoencoder CLIP:** The model concurrently reconstructs facial expressions while aligning their latent representation with corresponding text and image representations in the CLIP space. **(e) 'Surprise' Emotion Word Cloud:** closest words to 'surprise' using Emolex [11] based on cosine similarity of emotion distribution. **(f) FEG model Comparison** for the proposed FEG baselines on an example prompt.

4 METHOD

4.1 Models

In this section, we propose several baseline models for the task of translating emotion descriptions into 3D facial expressions. This

includes (i) fine-tuning of pre-trained language models, (ii) CLIP-based approaches, and (iii) Emotion-XLM, an architecture we have developed to enhance LM functionality.

Pretrained LM Baselines: We utilize BERT [4] and Glot500, a highly multilingual variant of XLM-RoBERTa [6], as the backbones. To map LM outputs into a designated target space, we incorporate

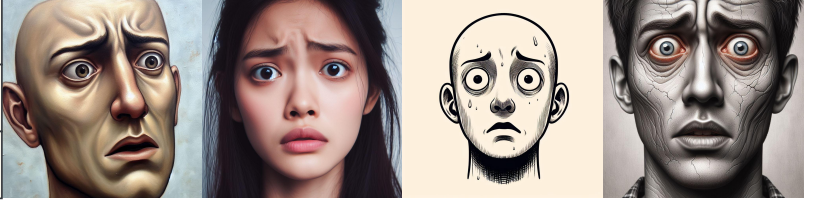
neutral	happiness	surprise	sadness	anger	disgust	fear	contempt
0.2	0	0	0.8	0	0	0	0
She had a round face with sunken cheeks, drooping eyelids and a furrowed brow that expressed sadness.							
The sadness she felt was evident in her face, her eyes were downcast and her cheeks had fallen inward.							
The emotion of sadness was clearly visible in her soft face, her eyebrows were drawn together and her eyes were heavy.							
neutral	happiness	surprise	sadness	anger	disgust	fear	contempt
0	1	0	0	0	0	0	0
The human face expresses happiness in a wide smile and sparkling eyes							
The person has a cheerful visage and a joyful countenance							
The person has a broad grin and eyes alight with cheer							
neutral	happiness	surprise	sadness	anger	disgust	fear	contempt
0.5	0	0.7	0.2	0	0	0.3	0
The human face emotion was one of total confusion, with wide eyes, slightly open mouth and furrowed brow.							
The human face expression was one of bewilderment, with eyes that were wide and a mouth slightly open, and furrowed brow.							
The human face emotion was one of bewilderment, with eyes that were wide and staring, a mouth slightly agape and a furrowed brow.							

Figure 2: Samples of Emo3d Dataset: A glimpse into the rich diversity and complexity of our collected data, paving the way for insightful analysis and discovery.

a Multi-Layer Perceptron (MLP). The MLP is trained with tuples $T = \{(b, l) \mid b \in \mathbb{R}^{768}, l \in \mathbb{R}^{52}\}$, where b denotes the LM output and l represents the corresponding blendshape scores.

Emotion-XLM: Extending the MLP structure to XLM-RoBERTa, we introduce an emotion-extractor unit. The transformer output is fed into this unit to extract emotion distributions alongside one-hot vectors. Representing the input space as $B = \{b \mid b \in \mathbb{R}^{768}\}$, the emotion-extractor unit produces output $E = \{(v, o) \mid v, o \in \mathbb{R}^8\}$, where v indicates emotion intensities in $V = \{[v_1, \dots, v_8] \mid v_i \in [0, 1], i = 1, \dots, 8\}$, and o is the one-hot vector of v . Pairs of vectors are then passed to the MLP unit, where they are concatenated with the text embedding before being fed to the regression unit, $\mathbb{F}(\cdot) : \mathbb{R}^{784} \rightarrow \mathbb{R}^{52}$. To train both parts efficiently, 50 % of

ground truth emotions are replaced with outputs from the emotion-extractor unit.

$$L = \lambda_1 L_{Blendshape} + \lambda_2 L_{Emotion} \quad (1)$$

Our training methodology employs a combination of MSE losses for blendshapes and extracted emotions, weighted by coefficients to balance their contributions effectively. This model is illustrated in Figure 1c.

CLIP Baseline: We employed a Multi-Layer Perceptron (MLP) structure on the CLIP [16] backbone. What distinguishes this model from Pretrained LM Baselines is the incorporation of both image

and text embeddings during training, effectively doubling the size of our dataset.

VAE CLIP We employed a Variational Autoencoder (VAE) to align blendshape scores with their corresponding text and image CLIP [16] embeddings, as illustrated in Figure 1d. The encoder maps blendshape scores to their respective text and image representations using cosine similarity, and the decoder sample from the latent space and reconstructs the blendshape scores. The reconstruction loss is defined by Mean Squared Error (MSE).

$$L_{text} = 1 - \cos(CLIP_{text}, z) \quad (2)$$

$$L_{image} = 1 - \cos(CLIP_{image}, z) \quad (3)$$

$$L = L_{recon} + \lambda_{text}L_{text} + \lambda_{image}L_{image} \quad (4)$$

Here, $\cos(a, b)$ denotes the cosine similarity between two vectors a and b .

Additionally, in the pursuit of a comprehensive comparison, we made diligent attempts to establish contact with the Expclip team through email and include the model in our comparison. Regrettably, as of the present moment, we have not received a response from them.

4.2 Emo3D Metric

We introduce a new 3D Facial Expression Generation (FEG) metric for evaluating the reconstruction of the original emotion vector from 2D snapshots of the generated 3D faces. We create a test set comprising diverse emotion prompts uniformly selected. To evaluate any proposed FEG model, we generate the corresponding blendshape scores of the input text and project the 3D face model onto a 2D image. Using zero-shot CLIP [16], we identify the k -nearest text prompts related to the image. We calculate the emotion distribution for the original prompt and the top- K prompts. This is followed by computing the Kullback-Leibler (KL) divergence between the emotion vector of the original prompt and the average emotion vector of the top- K retrieved prompts. We refer to the normalized KL bounded between 0 and 1 as the “Emo3D metric”:

$$D_{KL}(\phi \parallel \bar{\psi}) = \sum_i \phi(i) \cdot \log \left(\frac{\phi(i)}{\bar{\psi}(i)} \right) \quad (5)$$

$$\text{Emo3D Metric} = \frac{1}{1 + e^{-D_{KL}(\phi \parallel \bar{\psi})}} \quad (6)$$

where ϕ represents the emotion distribution of the input prompt, and $\bar{\psi}$ represents the mean emotion distribution of the top- k prompts. The steps for Emo3D calculation are outlined in Figure 1b. In our evaluation of the FEG models, we provide both the Emo3D Metric and the MSE scores of the 3D models for comparison purposes.

5 RESULTS

The FEG model performances are provided in Table 2. It becomes evident that the CLIP With Regression Unit model demonstrates superior performance when evaluated using our Emo3D metric. Our results indicate that the MSE and Emo3D metrics do not consistently align. When we examined the 3D model outputs, we observed that samples that performed better according to Emo3D metric also demonstrated a closer visual resemblance to the input prompt, in contrast to samples that showed better performance based on MSE,

Model	MSE	Emo3D
BERT	0.03	0.796
XLmRoBERTa	0.04	0.789
Autoencoder CLIP	0.002	0.776
Emotion-XLM	0.035	0.756
CLIP	0.014	0.737

Table 2: Performance comparison of FEG models using MSE vs. Emo3D metrics.

similar to Figure 1f. This can be because in our metric, Emo3D prioritizes visual-text alignment and proximity, tending to capture richer semantic information than distance metrics in 3D space using MSE.

6 CONCLUSION

In this paper, we introduced ‘Emo3D,’ a comprehensive ‘Text-Image-Expression dataset’ that covered a wide range of human emotions and their textual descriptions, paired with images and 3D blendshapes. Our use of Language Models (LLMs) to generate prompts captured a variety of emotional expressions and descriptions. To the best of our knowledge, ‘Emo3D’ stood out as the most comprehensive FEG dataset, encompassing sufficiently diverse and complex emotional descriptions. Furthermore, we developed an efficient evaluation metric to provide 3D image synthesis models with a reliable benchmark. Throughout our work, we tested several unimodal and multimodal models as baselines to encourage new entrants to the field. The significance of ‘Emo3D’ lay in its potential to advance 3D facial expression synthesis, holding promising implications for animation, virtual reality, and emotional human-computer interaction.

7 LIMITATIONS AND FUTURE WORK

While our dataset exhibits positive attributes, it is not without errors stemming from the processes involved in its production. Specifically, the use of Mediapipe to obtain blendshape scores introduced inaccuracies, particularly in the representation of certain emotions and facial expressions. To enhance the dataset in future endeavors, collaboration with skilled animators could be sought to refine and design more accurate blendshape scores.

8 ETHICS

This paper leverages GPT-3.5 [13] for generating textual emotional descriptions and DALL-E3 [17] for creating corresponding images. It’s vital to recognize the potential biases and privacy concerns inherent in these AI models. Both GPT-3.5 and DALL-E3, like many advanced AI systems, reflect the data on which they were trained, which can include societal biases and inaccuracies. Mitigating and analyzing such biases is beyond the scope of this paper. Studies such as “DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generation Models” [2] and the paper “ChatGPT: A Comprehensive Review on Background, Applications, Key Challenges, Bias, Ethics, Limitations and Future Scope” [18] thoroughly examine biases in the DALL-E and GPT models, respectively. According to these studies, the DALL-E and GPT models are also

shown to have certain degrees of biases related to gender, skin tone, professions, and certain attributes, and may have privacy or accountability concerns.

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